EVALUATION STUDY · IFFAR SÃO BORJA · INF/UFRGS

# Collab2Us: from visual ideation to an agile backlog, with human curation

Pilot study with teachers — study documentation (English version)

| | |
|---|---|
| **Research team** | Pedro Lucas Dornelles and Rafael Parizi — Federal Institute Farroupilha (IFFar), São Borja campus<br>Letícia Machado — Institute of Informatics, Federal University of Rio Grande do Sul (INF/UFRGS) |
| **Principal investigator** | Prof. Dr. Rafael Baldiati Parizi · rafael.parizi@iffar.edu.br |
| **Ethics approval (CAAE)** | 89811725.0.0000.5574 |
| **Study website** | https://rafaelparizi.github.io/collab2us-estudo/en/ |
| **Tool** | https://collab2us.onrender.com |

The study is conducted in Portuguese. The official consent form, approved by the Research Ethics Committee (CEP), is the Portuguese version; the English TAM3 and SUS items follow the original wording of the instruments, adapted to Collab2Us.

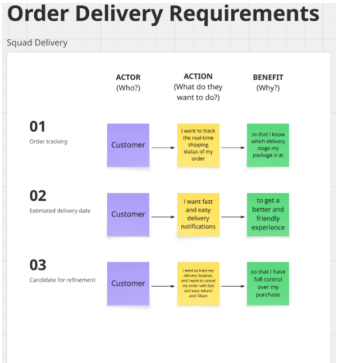
## 1. Study overview

In requirements workshops, stakeholders capture ideas on visual boards such as Miro, and the Product Owner has to turn them into well-scoped user stories in a backlog, such as Trello. Done by hand, this hand-off loses context; done only with generative AI, it requires checking whether the suggestions match what stakeholders expected. **Collab2Us** supports this transition in four steps. The tool proposes; the Product Owner decides.

| | |
|---|---|
| **0** | **Integration setup:** connect Miro and Trello and authorize the boards (done in advance by the researcher). |
| **1** | **Extraction:** read the board's notes and connectors and build candidate stories. |
| **2** | **Quality and curation:** QUS and INVEST support the review; the Product Owner edits and selects; optional AI suggests rewordings, splits and BDD criteria. |
| **3** | **Export and traceability:** stories become Trello cards linked to their source notes. |

**Scenario.** Ana, Product Owner of an online store, brings the Miro board *Order Delivery Requirements* to the backlog. It holds three candidate stories: order tracking (US01), delivery notifications labeled as “estimated delivery date” (US02), and a note that mixes location tracking, cancellation and refunds and ends unfinished (US03).

The indicators help flag problems of form; deciding what each story means and how to split US03 is up to Ana.




## 2. Session protocol

Individual sessions of about 30 to 35 minutes, with fictitious data and test accounts. Participants accept the consent form on the study website and follow the instructions page. They can ask for help, take a break or stop at any time.

1. **Brief introduction** to the tool and the flow; the session covers steps 1 to 3.
2. **Five activities as Product Owner:** extract ideas from Miro into candidate stories; examine the quality assessment (QUS and INVEST); revise a story with AI support and decide on the final text; export it to Trello and check the export report; follow the source link back to the original Miro note.
3. **Questionnaires:** 14 acceptance statements adapted from TAM3 and the 10 SUS statements.
4. **Three open questions**, audio-recorded only with the participant's specific authorization.

Completed questionnaires are sent to a spreadsheet owned by the principal investigator.

# 3. Informed Consent Form (translation)

> English translation for information. The official version, approved by the Research Ethics Committee (CEP) of the Federal Institute Farroupilha, is the Portuguese one.

**Pilot study of the Collab2Us tool with teachers.** Principal investigator: Prof. Dr. Rafael Baldiati Parizi, Federal Institute Farroupilha, São Borja campus (rafael.parizi@iffar.edu.br). Collaborating researcher: Pedro Lucas Dornelles dos Santos. CAAE: 89811725.0.0000.5574.

**1. Introduction and invitation.** You are invited to take part voluntarily in a pilot study of Collab2Us. Read this form and ask any questions you wish before deciding. The study evaluates the tool and the research procedure; it does not evaluate your competence or your professional performance. Declining or withdrawing will not harm your activities, professional relationships or institutional affiliation.

**2. Purpose of the research.** The purpose is to identify usability difficulties and opportunities to improve Collab2Us, and to check whether the tasks and instruments are suitable for a later study with students. The tool supports turning ideas from a Miro board into user stories, with quality review, artificial intelligence refinement and export to a Trello backlog, keeping links to the source of the information.

**3. Procedures and duration.** The pilot involves three teachers, in individual sessions of approximately 30 to 35 minutes. After consent, the researcher will briefly present the tool and an explanatory image. You will carry out six activities with fictitious data: extract ideas from Miro; examine the quality assessment; revise a story with AI support; export it to Trello; check its source; and verify a change or divergence between the records.

The researcher will note the progress of the tasks, the time, difficulties, errors and any requests for help. Afterwards, you will respond to 14 statements about acceptance of the tool, adapted from TAM3 constructs, and to the 10 statements of the SUS usability scale. Finally, three open questions will be asked about contributions, difficulties and improvements. You may decline to answer any question, ask for a break or end the session.

The interview may be recorded in audio only, subject to the specific authorization at the end of this form and to confirmation before recording starts. Declining audio recording does not prevent participation; in that case, answers will be recorded in writing. This form does not authorize video or screen recording.

**4. Risks and safeguards.** Taking part may cause tiredness, frustration with tool failures or discomfort when reporting difficulties. There is also a risk of undue exposure of information, especially because a voice can identify a person. To reduce these risks, fictitious data and test accounts, codes in the records and restricted access to the data will be used. You will not need to enter real personal or professional information into Miro, Trello or the AI service.

You may ask for help, a break or to end the session. The researcher will stop the recording whenever requested. If discomfort or harm related to your participation occurs, contact the principal investigator for support and appropriate action, without waiving any rights to assistance or compensation.

**5. Benefits and expenses.** There is no guarantee of direct benefit. Your participation may help improve the tool and the study with students. There will be no payment for participating and no charge for using the tool. Any expenses needed to take part should be discussed with the principal investigator, to ensure they do not fall on you.

**6. Privacy and use of results.** Records will be treated confidentially. Forms, observations and audio files will receive a participant code. The signed form and the list linking codes to identities will be stored separately. Voice and signature are identifiable; therefore, data collection is not anonymous. Even without names, accounts from a small group may allow indirect recognition, and the team will reduce details that could favor such identification.

The data will be used for scientific and educational purposes, including the undergraduate thesis (TCC), papers and presentations about Collab2Us. Aggregated results and de-identified text excerpts, reviewed to remove personal information, may be published. Names, signatures and audio files will not be published. Individual forms and recordings will not be part of a public artifact package. The SUS analysis will receive only the ten numeric answers, without name or contact details.

**7. Storage and withdrawal of data.** Access to the records will be restricted to the principal investigator and the authorized collaborator. After collection, the records will be kept in secure storage under the investigator's responsibility for five years after the end of the research. At the end of that period, they will be securely deleted. Temporary copies on recording devices or collection platforms will be removed after the files are transferred and checked.

You may withdraw your consent and request deletion of identifiable records through the investigator's contact. When the data have already been aggregated or irreversibly anonymized, it may not be possible to locate your contribution. Withdrawal will not allow removing results that have already been published. These conditions will be explained to you when you make a request.

**8. Questions and copy of the form.** You may ask for clarification before, during or after taking part, using the investigator's contact details above. For questions or complaints about ethical aspects, the institutional template indicates the Research Ethics Committee of the Federal Institute Farroupilha, email cep@iffar.edu.br.

This form may be signed in two paper copies, one for you and one for the researcher, or electronically, including through gov.br, with a signed copy delivered to you. Data collection will only begin after your consent. Authorization for audio recording will be confirmed again before the interview.

**9. Statement of consent.** I declare that I have read this form, or that it was read to me, that I received explanations about the objectives, procedures, risks and benefits, and that I had the opportunity to clarify my questions. I know that I may decline questions, stop the session or withdraw my consent without any harm. I will receive a copy of this form. By signing, I voluntarily agree to take part in the study and authorize the use of the data under the conditions described.

**Specific authorization for audio recording** (independent of the agreement to take part):

☐ I **AUTHORIZE** audio recording of my answers to the three open questions, solely for recording, transcription and analysis in the research, under the conditions of this form. I know I can ask for the recording to be stopped.

☐ I **DO NOT AUTHORIZE** audio recording. My answers may be recorded in writing by the researcher.

Participant's name: _____ Place and date: _____, ____/____/_____

Participant's signature: _____

I declare that I presented the study information, clarified any questions and will respect the participant's choices and rights described in this form.

Researcher who obtained consent: _____ Signature: _____ Date: ____/____/_____

# 4. Questionnaire

Participant code: _____ Date: ____/____/_____

Three parts: 14 statements about acceptance, 10 about usability and three open questions. There are no right or wrong answers; respond based on the tasks you carried out.

## Part 1 · Acceptance (adapted from TAM3)

"Requirements tasks" means turning ideas from the visual board into user stories that are reviewed and organized in the backlog.

1 = Strongly disagree · 2 = Moderately disagree · 3 = Somewhat disagree · 4 = Neither agree nor disagree · 5 = Somewhat agree · 6 = Moderately agree · 7 = Strongly agree

| | | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|---|
| **PU1** | Using Collab2Us improves my performance in requirements tasks. | | | | | | | |
| **PU2** | Using Collab2Us in requirements tasks increases my productivity. | | | | | | | |
| **PU3** | Using Collab2Us enhances my effectiveness in requirements tasks. | | | | | | | |
| **PU4** | I find Collab2Us to be useful in requirements tasks. | | | | | | | |
| **PEOU1** | My interaction with Collab2Us is clear and understandable. | | | | | | | |
| **PEOU2** | Interacting with Collab2Us does not require a lot of my mental effort. | | | | | | | |
| **PEOU3** | I find Collab2Us to be easy to use. | | | | | | | |
| **PEOU4** | I find it easy to get Collab2Us to do what I want it to do. | | | | | | | |
| **BI1** | Assuming I had access to Collab2Us, I intend to use it. | | | | | | | |
| **BI2** | Given that I had access to Collab2Us, I predict that I would use it. | | | | | | | |
| **BI3** | I plan to use Collab2Us in the next three months. | | | | | | | |
| **ENJ1** | I find using Collab2Us to be enjoyable. | | | | | | | |
| **ENJ2** | The actual process of using Collab2Us is pleasant. | | | | | | | |
| **ENJ3** | I have fun using Collab2Us. | | | | | | | |

*For BI items, assume you have access to Collab2Us and opportunities to carry out requirements tasks. For ENJ items, consider the experience of using the tool, regardless of its usefulness.*

## Part 2 · Usability (SUS)

1 = Strongly disagree · 2 = Disagree · 3 = Neither agree nor disagree · 4 = Agree · 5 = Strongly agree

| | | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| **SUS1** | I think that I would like to use Collab2Us frequently. | | | | | |
| **SUS2** | I found Collab2Us unnecessarily complex. | | | | | |
| **SUS3** | I thought Collab2Us was easy to use. | | | | | |
| **SUS4** | I think that I would need the support of a technical person to be able to use Collab2Us. | | | | | |
| **SUS5** | I found the various functions in Collab2Us were well integrated. | | | | | |
| **SUS6** | I thought there was too much inconsistency in Collab2Us. | | | | | |
| **SUS7** | I would imagine that most people would learn to use Collab2Us very quickly. | | | | | |
| **SUS8** | I found Collab2Us very cumbersome to use. | | | | | |
| **SUS9** | I felt very confident using Collab2Us. | | | | | |
| **SUS10** | I needed to learn a lot of things before I could get going with Collab2Us. | | | | | |

## Part 3 · Open questions

Answer in your own words and, whenever possible, give a concrete example.

**Q1.** Which part of Collab2Us helped you most in moving the ideas into the backlog?

**Q2.** Which part did you find most difficult, or where were you unsure what to do?

**Q3.** What needs to change before we use the tool with students?

**References.** Brooke, J. (1996). SUS: A “quick and dirty” usability scale. · Lucassen, G. et al. (2016). The Quality User Story framework and tool. *Requirements Engineering*, 21. · Venkatesh, V.; Bala, H. (2008). Technology Acceptance Model 3. *Decision Sciences*, 39(2). · Wake, B. (2003). INVEST in Good Stories, and SMART Tasks.